

MANUSCRIPT ID: JBI-25-1179

MANUSCRIPT TITLE: A LangChain-Based Pipeline for One-Shot Synthetic Text Generation Using Generative Pre-Trained Transformers in Palliative Care Research

The authors would like to thank the reviewers for their valuable feedback on this manuscript. Below we highlight our point-by-point response to all the comments made.

REVIEWER 1 COMMENT	AUTHOR RESPONSE
1. It is necessary to provide additional explanation for why the parameter "25 times completion x 25 notes" was selected (Section 3.3.2). Is there any experimental evidence to prove that this configuration is optimal?	We have added additional explanation for this parameter selection. A more detailed explanation is now provided in Section 3.3.2 on lines 243 to 255.
2. It should be clarified how to handle cases of generation failure or formatting errors (Section 4.1 mentions "minor data processing issues").	In Section 4.1, the lines 348 to 350 and lines 352 to 359 have been added to clarify this error handling.
3. The generalization ability of the prompt templates in Figures 2 and 3 needs to be verified. Suggest adding comparative experiments with different prompt styles	Thank you for this suggestion. We have updated Section 3.4.2, lines 327 to 336, and Section 4.3.7, lines 457 to 476, to add some prompt comparison experiments.
4. Reasons why GPT-3.5 is superior to GPT-4o? The current explanation (regarding differences in training data or complexity issues) is too speculative. Suggest verifying through controlled experiments (such as testing different models with fixed prompts).	Further clarification has been added to Section 5 on lines 503 to 505 to reduce the speculative nature of results. Additionally, we added temporal experimentation to our work to show the speed of the GPT-3.5 model in comparison to the GPT-4o model. These temporal experiments can be found in Section 3.4.2, lines 320 to 326, Section 4.3.6, lines 452 to 455, and Section 5, lines 505 to 508.
5. Only one expert participating in the evaluation may introduce bias (Section 5.3). It is recommended that at least 3 experts independently annotate and calculate inter group consistency (such as Fleiss' Kappa).	We greatly appreciate this feedback; we subsequently carried out a 3-person evaluation to strengthen our results. Section 3.4.1, lines 278 to 288, has been updated to reflect a 3-person evaluation. Additionally, the entirety of Section 4.2 has been updated to reflect our 3-person test results.
6. Potential differences between generated notes and actual records	These notes do not heavily use clinical terminology, as they are more so to do with

need to be discussed, such as emotional expression and accuracy of clinical terminology.	every-day care as opposed to diagnosis of medical conditions. As a result, they differ enormously from the sorts of notes found in hospitals and other clinical practices. We have added additional clarification to Section 4.2.3, lines 399 to 401, to reflect the general tone found in the notes as assessed by multiple members of our research team.
7. Despite using fictional inputs, it is still necessary to analyze whether the GPT model may remember and leak sensitive information from the training set.	As we do not have direct access to the GPT model training data, we cannot analyze the leakage level of this model; however, we have added additional clarification about our access rights and future directions for research to Section 5, lines 548 to 557, with this feedback in mind.
8. Merge "Future Directions" (Section 5.2) and "Limitations" (Section 5.3) into a "Discussion" subsection to avoid duplication.	These sections have been merged into Section 5 to avoid duplication.
9. Supplement some recent research on synthetic data generation.	Recent research on synthetic data generation has been added to Section 2, lines 124 to 137.
REVIEWER 2 COMMENT	AUTHOR RESPONSE
test the temperature/context-window-size parameters while using the GPT model... it is worth testing if the configuration of these models impacted the results	The context-window size parameter is fixed and only changes depending on which model you use. We have added some additional clarification surrounding this parameter to Section 3.1, lines 176 and 180. The temperature parameter was based on past literature. Adjusting it will either render results too random or too consistent. Additional justification for our choice of temperature has been added to Section 3.1, lines 184 to 189.